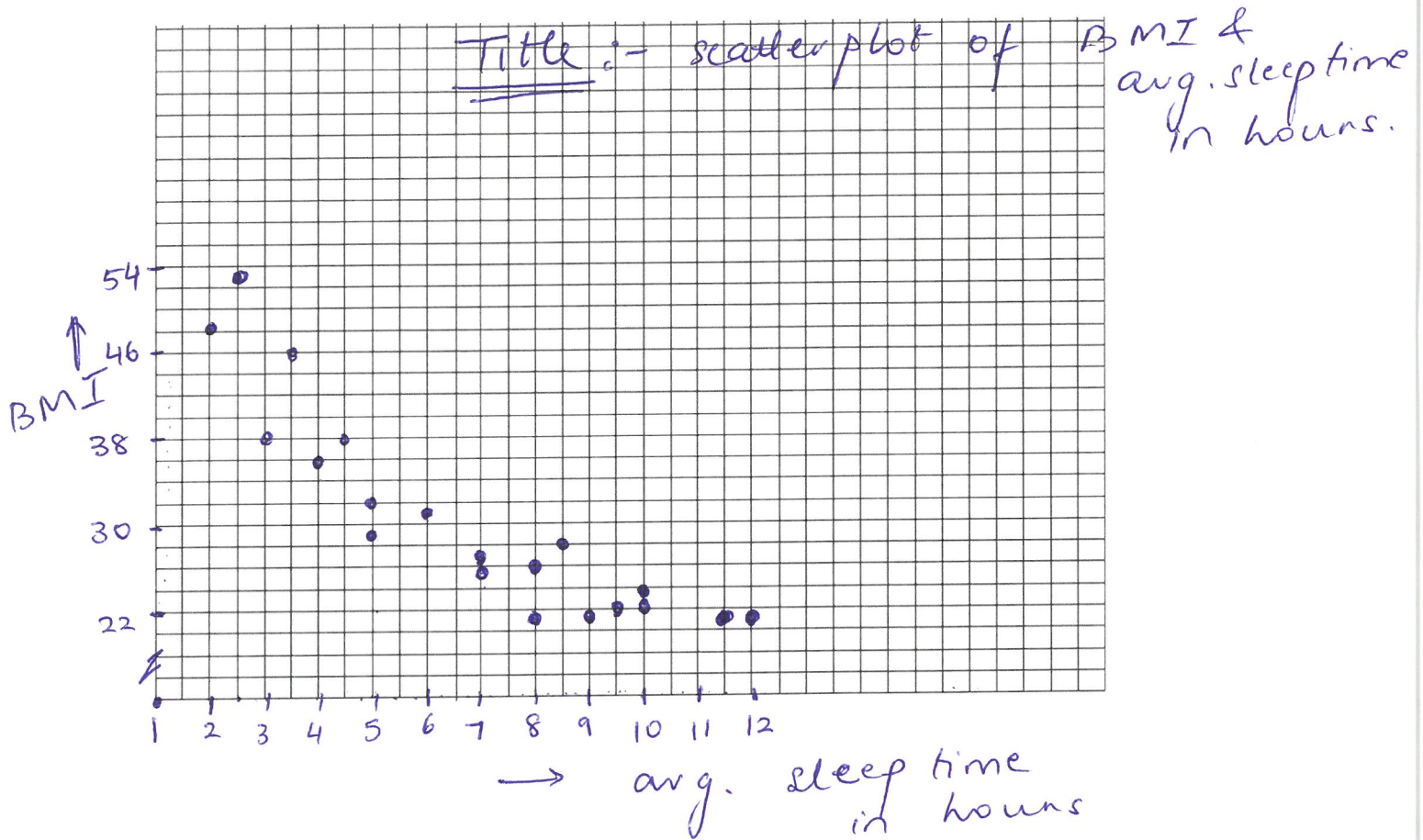


Answers i-vi

i) Scatterplot

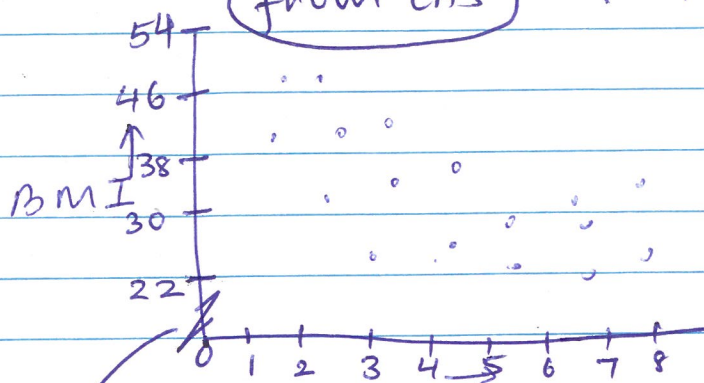


Discussion and working out for ii-iv

Data AnalysisSAC RevisionPart B

(i)

Title :- Scatterplot for Body Mass Index
from CAS & Average daily sleep time (hours)



The scatterplot has a non-linear pattern/trend between the 2 variables BMI & sleep.

Do not forget this

avg. sleep time in hours

Labels

(ii)

Regression Analysis with original data

a) $BMI = 50.06 - 2.927(\text{sleep hours})$

b) Interpret - y-intercept $\rightarrow (50.06)$

The BMI = 50.06 when the average daily sleep time (hours) is zero.

So with no sleep the BMI = 50.06

Interpret gradient $\rightarrow (-2.927)$

As the avg. daily sleep time increases by 1 hour, the BMI reduces by 2.927

(c)

coefficient of correlation $\rightarrow r = -0.894$

The value of 'r' indicates a

negative

correlation between the BMI & average daily sleep time (hours)

The association is strong.

However, due to non-linearity of the data, it is not very reliable calculation. as regression has an underlying assumption

(d) coefficient of determination $(r^2) = .7995$

Interpretation :- 79.95% of the variation in BMI can be explained by variation in avg. no. of sleep time (hours)

This is again not a very reliable calculation due to Non-Linearity of the data distribution.

(iii) Avg. sleep time in hours of a student selected at random = 7 hrs.

Equation :-

predicted

$$BMI = 50.06 - 2.927(7)$$

$$= \cancel{29.56468}$$

$$= 29.5646$$

use actual values from CAS for further calculation
(rounded fig)

Actual when sleep time = 7 hrs

$\downarrow$ BMI
(from data)

Actual BMI = 25

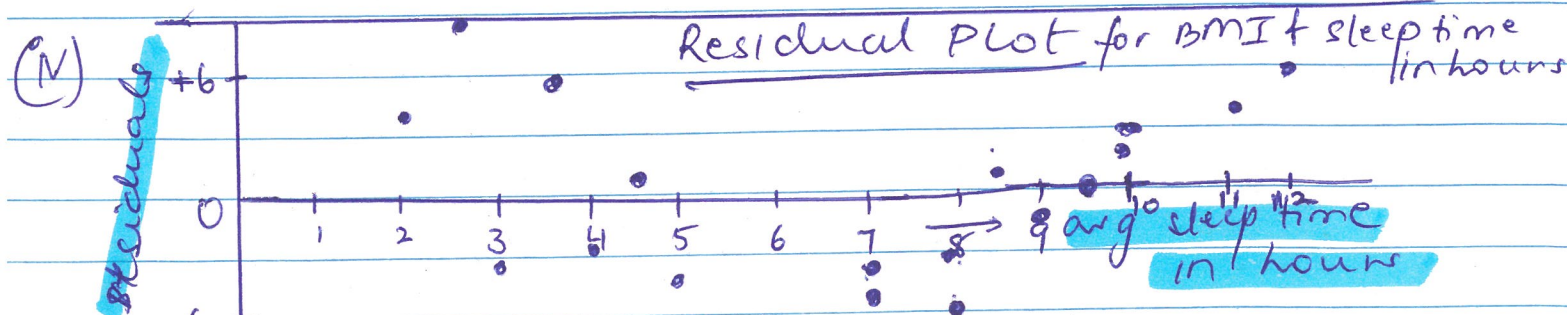
These are rounded figures.

Residual = Actual - Predicted

$$= 25 - 29.5646$$

$$= -4.5646$$

If the question says round to 4 significant figures



(v) Based on the shape of the scatter plot, the following transformations can be done.

$$\frac{1}{x}, \log x, \log y, \frac{1}{y}$$

choose any two

(1) Log x transformation :-

Equation ($y = a + bx$)

where
 $y = \text{BMI}$
 $x = \log x$

$$\text{BMI} = 62.50 - 41.38 (\log x)$$

$$\text{BMI} = 62.50 - 41.38 (\log (\text{sleep time}))$$

$$r^2 = 0.8936 = 89.36\%$$

(2) $\frac{1}{y}$ transformation

$$y = \frac{1}{\text{BMI}} \quad (\text{rec BMI})$$

$$\frac{1}{\text{BMI}} = 0.01529 + 0.003071 (\text{sleep time})$$

$x = \text{sleep hours time}$

$$r^2 = 0.8878 = 88.78\%$$

The Log 'x' transformation will be best suited due to higher r^2 of 89.36% compared to $\frac{1}{y}$ transformation with $r^2 = 88.78\%$

(VI) Effective least square regression line

$$BMI = 62.50 - 41.38 (\log \text{ sleep time})$$

BMI of 31

solve

$$31 = 62.50 - 41.38 (\log \text{ ~~sleep time~~ }^x, x)$$

$$x = 5.77958$$

These are rounded figures

Use complete values not rounded off figures

when sleep time = 3 hrs

$$42.772$$

predicted BMI =

$$\rightarrow 62.4952 - 41.3379 (\log_{10} 3) = 42.772$$

These are what actual figures look like

The predictions are reliable because it is interpolated data - lies within the actual data range.